IOP LAB ASSIGNMENT 3 and 4

27. PROGRAM TO CHECK WHETHER A NO IS PRIME OR NOT.

```c
#include<stdio.h>
int main()
{
    int i, n, flag ;
    printf("Enter the number");
    scanf("%d", &n);
    for(i=2;i<=n/2;i++)
        {
            if(n%i==0)
            {
                flag=1;
                break;
            }
            else
            {
                flag=0;
            }
        }
    if(flag==0)
    {
        printf("Prime Number");
    }
            else
    {
        printf("No prime Number");
    }
            return 0;
}
```

```
Enter the number13
Prime Number
```

28. PROGRAM TO CHECK WHETHER A NO IS ARMSTRONG OR NOT.

```c
#include<stdio.h>
#include<math.h>
int main()
{
    int n, temp, digit, numdigit=0;
    double sum=0;
    printf("Enter the number:");
    scanf("%d", &n);
    temp = n;
    while(temp!=0)
    {
        temp=temp/10;
        numdigit++;
    }
    temp=n;
    while(temp!=0)
    {
        digit=temp%10;
        sum+=pow(digit, numdigit);
        temp =temp/10;
    }
if(sum==n)
{
    printf("ARMSTRONG");
}
    else
{
    printf("NO ARMSTRONG");
}
    return 0;
}
```

```
Enter the number:153
ARMSTRONG
```

29. PROGRAM TO CHECK WHETHER A NO IS PALINDROM OR NOT.

```c
#include<stdio.h>
int main()
{
    int n, digit, temp, reverse=0, sum=0;
    printf("Enter the number:");
    scanf("%d", &n);
    temp=n;
    while(temp!=0)
        {
            digit=temp%10;
            reverse=reverse*10+digit;
            temp=temp/10;

        }
    if(reverse==n)
    {
        printf("PALINDROME");
    }
    else
    {
        printf("NO PALINDROME");
    }
    return 0;
}
```

```
Enter the number:12221
PALINDROME
```

30. PROGRAM TO REVERSE OF A GIVEN NO.

```c
#include<stdio.h>
int main()
{
    int n;
    printf("Enter the number:");
    scanf("%d", &n);
    while(n!=0)
    {
        printf("%d", n%10);
        n=n/10;
    }
    return 0;

}
```

```
Enter the number:345
543
```

31. PROGRAM TO PRINT SUM OF INDIVIDUAL DIGITS OF A ‘N’ DIGIT NO.

```c
#include<stdio.h>
int main()
{
    int n, sum=0;
    printf("Enter the number:");
    scanf("%d", &n);
    while(n!=0)
        {
            if(sum!=0)
            {
                printf("+");
            }
            printf("%d", n%10);
            sum+=n%10;
            n=n/10;
        }
    printf("=%d", sum);
    printf("\nThe sum of digits is %d", sum);
    return 0;
}
```

```
Enter the number: 34563
36543
```

32. PROGRAM TO FINDOUT MAX AND SECOND MAX FROM ‘N’ NOS .

```c
#include <stdio.h>
#include <limits.h>

int main() {
    int n, i, num;
    int max = INT_MIN, second = INT_MIN;

    printf("Enter how many numbers: ");
    scanf("%d", &n);

    if (n < 2) {
        printf("Enter at least 2 numbers.\n");
        return 0;
    }

    printf("Enter %d numbers:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &num);

        if (num > max) {
            second = max;      // old max becomes second max
            max = num;
        } else if (num > second && num != max) {
            second = num;
        }
    }

    if (second == INT_MIN)
        printf("Maximum = %d\nNo second maximum (all numbers are equal).\n", max);
    else
        printf("Maximum = %d\nSecond Maximum =%d\n", max, second);

    return 0;
}
```

```
Enter how many numbers: 5
Enter 5 numbers:
23 45 55 66 77
Maximum = 77
Second Maximum =66
```

33. PROGRAM TO FINDOUT MAX AND MIN FROM ‘N’ NOS.

```c
#include <stdio.h>

int main() {
    int n, i, num, max, min;

    printf("Enter how many numbers: ");
    scanf("%d", &n);

    if (n < 1) {
        printf("Enter at least 1 number.\n");
        return 0;
    }

    printf("Enter %d numbers:\n", n);
    scanf("%d", &num);
    max = min = num;           // first number is both max and min

    for (i = 1; i < n; i++) {
        scanf("%d", &num);

        if (num > max)
            max = num;
        if (num < min)
            min = num;
    }

    printf("Maximum = %d\n", max);
    printf("Minimum = %d\n", min);

    return 0;
}
```

```
Enter how many numbers: 5
Enter 5 numbers:
12 34 22 33 45
Maximum = 45
Minimum = 12
```

34. PROGRAM TO PRINT FIBONACCI SERIES.(0,1,1,2,3...)

```c
#include <stdio.h>

int main() {
    int n, i;
    long long a = 0, b = 1, next;

    printf("Enter number of terms: ");
    scanf("%d", &n);

    if (n < 1) {
        printf("Enter at least 1 term.\n");
        return 0;
    }

    printf("Fibonacci series: ");
    for (i = 1; i <= n; i++) {
        printf("%lld ", a);
        next = a + b;
        a = b;
        b = next;
    }
    printf("\n");

    return 0;
}
```

```
Fibonacci series: 0 1 1 2 3 5 8 13 21 34
```

35. READ A NUMBER N AND PRINT A SINGLE DIGIT ANSWER SHOWING SUM OF DIGIT OF N

```c
#include <stdio.h>

int main() {
    int n, sum;

    printf("Enter a number: ");
    scanf("%d", &n);

    if (n < 0)
        n = -n;

    while (n > 9) {
        sum = 0;
        while (n > 0) {
            sum += n % 10;
            n /= 10;
        }
        n = sum;
    }

    printf("Single digit sum = %d\n", n);

    return 0;
}
```

```
Enter a number: 56
Single digit sum = 2
```

36. PROGRAM TO PRINT 1,3,5,7,9.....N.

```c
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter t
    scanf("%d", &n)

    printf("Odd nur
    for (i = 1; i <
        printf("%d
    }
    printf("\n");

    return 0;
}
```

```
Enter the value of N: 35
Odd numbers: 1 3 5 7 9 11 13 15 17 19 21 23 25 27 29 31 33 35
```

37. PROGRAM TO PRINT 2,4,6,8,10,12.....N.

```c
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter the
    scanf("%d", &n);

    printf("Even numbe
    for (i = 2; i <= n
        printf("%d ",
    }
    printf("\n");

    return 0;
}
```

```
Enter the value of N: 34
Even numbers: 2 4 6 8 10 12 14 16 18 20 22 24 26 28 30 32 34
```

38. PROGRAM TO PRINT 1,4,9,16,25,.....N.

```c
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter the value of N: ");
    scanf("%d", &n);

    printf("Squares: ");
    for (i = 1; i * i <= n; i++) {
        printf("%d ", i * i);
    }
    printf("\n");

    return 0;
}
```

```
Enter the value of N: 42
Squares: 1 4 9 16 25 36
```

39. PROGRAM TO PRINT 1/1!+2/2!+3/3!+4/4!,.....N.

```c
int main() {
    int n = 0, i;
    double fact = 1, sum = 0;

    printf("Enter the value of N: ");
    if (scanf("%d", &n) != 1 || n < 1 || n > 20) {
        printf("Enter a whole number from 1 to 20.\n");
        return 0;
    }

    for (i = 1; i <= n; i++) {
        fact *= i;
        sum += i / fact;
    }

    printf("Sum = %.4f\n", sum);

    return 0;
}
```

```
nter the value of N: 13
um = 2.7183
```

40. READ NUMBERS TILL A NEGATIVE NUMBER IS ENTERED AND CALCULATE SUM OF A LIST OF NUMBERS READ.

```c
#include <stdio.h>

int main() {
    int num;
    long long sum = 0;

    printf("Enter numbers (negative number to stop):\n");

    while (scanf("%d", &num) == 1 && num >= 0) {
        sum += num;
    }

    printf("Sum = %lld\n", sum);

    return 0;
}
```

```
Enter numbers (negative number to stop):
12 34 56 45 -67
Sum = 147
```

41. READ A NUMBER N AND PRINT FACTOR OF N.(EX 28= 1,2,4,7,14,28)

```c
#include <stdio.h>

int main() {
    int n = 0, i;

    printf("Enter a number: ");
    if (scanf("%d", &n) != 1 || n < 1) {
        printf("Enter a positive whole number.\n");
        return 0;
    }

    printf("Factors of %d: ", n);
    for (i = 1; i <= n; i++) {
        if (n % i == 0) {
            printf("%d ", i);
        }
    }
    printf("\n");

    return 0;
}
```

```
Enter a number: 78
Factors of 78: 1 2 3 6 13 26 39 78
```

42. READ A NUMBER N AND PRINT IF N IS PRIME OR COMPOSITE.

```c
#include<stdio.h>
int main()
{
    int n, flag, i;
    printf("enter the number:");
    scanf("%d", &n);
    for(i=2;i<=n/2;i++)
        {
            if(n%i==0)
            {
              flag=1;
                break;
            }
            else
            {
                flag=0;
            }

        }
    if(flag==0)
    {
        printf("PRIME");
    }
    else
    {
        printf("COMPOSITE");
    }
    return 0;
}
```

```
enter the number:23
PRIME
```

43. WRITE A C PROGRAM TO CALCULATE THE COMMISSION FOR A SALES REPRESENTATIVE AS PER THE SALES AMOUNT GIVEN BELOW. If sales<=Rs. 500, commission is 5% If sales> but <=2000, commission is Rs. 35 plus 10% above Rs. 500 If sales>2000 but <=5000, commission is Rs. 185 plus 12% above Rs. 2000 If sales>5000, commission is 12.5%

```c
#include <stdio.h>

int main() {
    float sales, commission;

    printf("Enter sales amount: ");
    scanf("%f", &sales);

    if (sales <= 500)
        commission = 0.05 * sales;
    else if (sales <= 2000)
        commission = 35 + 0.10 * (sales - 500);
    else if (sales <= 5000)
        commission = 185 + 0.12 * (sales - 2000);
    else
        commission = 0.125 * sales;

    printf("Commission = Rs. %.2f\n", commission);
    return 0;
}
```

```
Enter sales amount: 4566
Commission = Rs. 492.92
```

44. WRITE A PROGRAM IN C FOR THE FOLLOWING. AN ELECTRIC DISTRIBUTION COMPANY CHARGES ITS DOMESTIC CONSUMERS AS FOLLOWS.

```c
#include <stdio.h>

int main() {
    float units, bill;

    printf("Enter units consumed: ");
    scanf("%f", &units);

    if (units <= 200)
        bill = 0.50 * units;
    else if (units <= 400)
        bill = 100 + 0.65 * (units - 200);
    else if (units <= 600)
        bill = 230 + 0.80 * (units - 400);
    else
        bill = 425 + 125 * (units - 600);

    printf("Amount to be paid = Rs. %.2f\n", bill);
    return 0;
}
```

```
Enter sales amount: 567
Commission = Rs. 41.70
```